

Credit Card Fraud Detection

Data Science / Machine Learning Project Report

Objective

The objective is to build a machine-learning pipeline that identifies potentially fraudulent credit-card transactions. The project focuses on severe class imbalance, resampling with SMOTE, supervised classification, and evaluation using fraud-sensitive metrics.

Dataset

The project is designed for the Kaggle Credit Card Fraud Detection dataset. The dataset contains transaction features and a target column named Class, where 0 represents a legitimate transaction and 1 represents fraud.

Methodology

- 1 Load and inspect the transaction dataset.
- 2 Measure the class imbalance between legitimate and fraudulent transactions.
- 3 Scale the Time and Amount features.
- 4 Split the data into stratified training and test sets.
- 5 Apply SMOTE only to the training set to avoid test-data leakage.
- 6 Train Logistic Regression, Random Forest and XGBoost classifiers.
- 7 Evaluate models with precision, recall, F1-score, ROC-AUC, PR-AUC and confusion matrices.
- 8 Save model comparison results, plots and trained models.

Why class imbalance matters

Fraud cases are much rarer than legitimate transactions. A model that predicts almost every transaction as legitimate can appear accurate while missing many fraud cases. Therefore, precision, recall, F1-score and precision-recall analysis are particularly useful alongside ROC-AUC.

Models

Model	Purpose
Logistic Regression	Baseline linear classification model.
Random Forest	Tree ensemble capable of learning nonlinear relationships.
XGBoost	Gradient-boosted decision trees for strong classification performance.

Evaluation Metrics

Precision: Of transactions predicted as fraud, how many are actually fraud.

Recall: Of actual fraud transactions, how many are detected.

F1-score: Harmonic mean of precision and recall.

ROC-AUC: Measures ranking/separation across classification thresholds.

PR-AUC: Particularly informative when the positive class is rare.

Confusion matrix: Shows true positives, true negatives, false positives and false negatives.

Important Implementation Detail

SMOTE is performed after the train-test split and only on training data. This prevents information from the test set from influencing synthetic training examples.

How to Run

1. Download creditcard.csv from Kaggle. 2. Put it inside the project's data folder. 3. Open the folder in VS Code. 4. Run `pip install -r requirements.txt`. 5. Run `python main.py`. 6. Review charts and model_comparison.csv in the outputs folder.

Conclusion

This project demonstrates a complete fraud-detection workflow from data inspection and imbalance analysis through preprocessing, SMOTE, model training, evaluation and model saving. The final model should be selected using the project's measured validation/test metrics and the practical cost of false positives versus false negatives.

Technologies: Python, Pandas, NumPy, Matplotlib, Scikit-learn, Imbalanced-learn, XGBoost, Joblib.